

## SQL Query Optimizer Agent

### PROBLEM STATEMENT

Database queries are often written inefficiently, leading to several performance and scalability challenges in modern applications. Common issues include:

- Slow execution times
- High database resource consumption
- Poor application performance
- Inefficient execution plans
- Scalability issues

The SQL Query Optimizer Agent is designed to help developers and database administrators analyze SQL queries, identify performance bottlenecks, understand execution plans, and generate optimized SQL recommendations using Artificial Intelligence. The system provides intelligent insights, query rewrites, index suggestions, and performance comparisons to improve overall database efficiency and application responsiveness.

**TEAM NAME -- Code Hawks**

### TEAM MEMBERS

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**Project Name :** SQL Query Optimizer Agent

**Team Name :** Code Hawks

**Domain :** Artificial Intelligence & Database Optimization

### Technology Stack:

React, FastAPI, Python, SQLite, Ollama, Tailwind CSS, Vite, Git, GitHub